

	$\mathcal{K}_{0^+}^{\#1}$	$\mathcal{K}_{0^-}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{K}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{K}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$3k^2{}^{(4)}\kappa_{15}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{K}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{K}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	0	0	0	0		
$\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	0	0	0	0		
$\mathcal{K}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	$-k^2{}^{(4)}\kappa_2$	$\frac{\gamma_1 k^2}{2\sqrt{2}}$		
$\mathcal{K}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	$\frac{\gamma_1 k^2}{2\sqrt{2}}$	$\frac{\gamma_2 k^2}{4}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{K}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
				$\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{3\gamma_3 k^2}{2}$	0
				$\mathcal{K}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	0

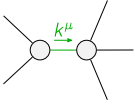
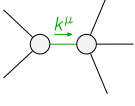
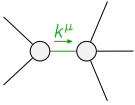
	$\mathcal{J}_{0^+}^{\#1}$	$\mathcal{J}_{0^-}^{\#1}{}_{\alpha\beta\chi}$				
$\mathcal{J}_{0^+}^{\#1}{}^\dagger$	0	0				
$\mathcal{J}_{0^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	$\frac{1}{3k^2{}^{(4)}\kappa_{15}}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha\beta}$	$\mathcal{J}_{1^+}^{\#1}{}_{\alpha}$	$\mathcal{J}_{1^+}^{\#2}{}_{\alpha}$
$\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	0	0	0	0		
$\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha\beta}$	0	0	0	0		
$\mathcal{J}_{1^+}^{\#1}{}^\dagger{}^{\alpha}$	0	0	$\frac{\gamma_2 k^2}{4\text{Det}(1^-)}$	$-\frac{\gamma_1 k^2}{2\sqrt{2}\text{Det}(1^-)}$		
$\mathcal{J}_{1^+}^{\#2}{}^\dagger{}^{\alpha}$	0	0	$-\frac{\gamma_1 k^2}{2\sqrt{2}\text{Det}(1^-)}$	$-\frac{k^2{}^{(4)}\kappa_2}{\text{Det}(1^-)}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta}$	$\mathcal{J}_{2^+}^{\#1}{}_{\alpha\beta\chi}$
				$\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta}$	$\frac{1}{\text{Det}(2^+)}$	0
				$\mathcal{J}_{2^+}^{\#1}{}^\dagger{}^{\alpha\beta\chi}$	0	0

Abbreviations used in matrices

$$\begin{aligned} \gamma_1 &== 2\kappa_2 + \kappa_7 \&\& \gamma_2 == 3\kappa_1 + \kappa_2 - 2\kappa_7 \&\& \gamma_3 == \kappa_1 + \kappa_2 \&\& \\ \text{Det}(1^-) &== -\frac{1}{8}k^4(6\kappa_2(\kappa_1 + \kappa_2) + \kappa_7^2) \&\& \text{Det}(2^+) == \frac{3}{2}k^2(\kappa_1 + \kappa_2) \end{aligned}$$

Lagrangian

$$\begin{aligned} &{}^{(4)}\kappa_1\partial_\beta\mathcal{K}^\delta{}_{\chi\delta}\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta + {}^{(4)}\kappa_2\partial_\chi\mathcal{K}^\delta{}_{\beta\delta}\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta - \frac{3}{2}{}^{(4)}\kappa_1\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\alpha\chi} + \\ &2{}^{(4)}\kappa_{15}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\alpha\chi} - \frac{3}{2}{}^{(4)}\kappa_2\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\alpha\chi} - 4{}^{(4)}\kappa_{15}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} - \\ &\frac{3}{2}{}^{(4)}\kappa_1\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\chi\alpha} - 2{}^{(4)}\kappa_{15}\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\chi\alpha} - \frac{3}{2}{}^{(4)}\kappa_2\partial_\beta\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\chi\alpha} + \\ &{}^{(4)}\kappa_7\partial^\chi\mathcal{K}^\alpha{}_\alpha{}^\beta\partial_\delta\mathcal{K}^\delta{}_{\chi\beta} - {}^{(4)}\kappa_{15}\partial_\alpha\mathcal{K}^{\alpha\beta\chi}\partial_\delta\mathcal{K}^\delta{}_{\beta\chi} + {}^{(4)}\kappa_{15}\partial_\delta\mathcal{K}^{\alpha\beta\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi} - 2{}^{(4)}\kappa_{15}\partial_\delta\mathcal{K}^{\alpha\beta\chi}\partial^\delta\mathcal{K}^{\alpha\beta\chi} \end{aligned}$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\mathcal{J}_{\alpha\beta\chi}$		
Source constraint(s)	# constraint(s)	Covariant form	
$\mathcal{J}_{0^+}^{\#1} == 0$	1	$\partial_\beta\mathcal{J}^\alpha{}_\alpha{}^\beta == 0$	
$\mathcal{J}_{1^+}^{\#2\alpha\beta} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\chi\alpha\beta} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta}$	
$\mathcal{J}_{1^+}^{\#1\alpha\beta} == 0$	3	$\partial_\delta\partial_\chi\partial^\alpha\mathcal{J}^{\chi\beta\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\beta\alpha\chi} == \partial_\delta\partial_\chi\partial^\beta\mathcal{J}^{\chi\alpha\delta} + \partial_\delta\partial^\delta\partial_\chi\mathcal{J}^{\alpha\beta\chi}$	
$\mathcal{J}_{2^+}^{\#1\alpha\beta\chi} == 0$	5	$3\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha\mathcal{J}^{\delta\beta}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\alpha\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\chi\alpha\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta\mathcal{J}^{\delta\alpha\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\beta\alpha\delta} +$ $4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\delta\alpha\beta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\alpha\beta\chi} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha\mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\beta\epsilon} + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\delta\mathcal{J}^{\delta\alpha}{}_\delta ==$ $3\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha\epsilon} + 3\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta\mathcal{J}^{\delta\alpha}{}_\delta + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\beta\chi\delta} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\chi\beta\delta} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha\mathcal{J}^{\delta\beta\chi} + 2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi\mathcal{J}^{\alpha\beta\delta} +$ $2\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\beta\alpha\chi} + 4\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta\mathcal{J}^{\chi\alpha\beta} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta\mathcal{J}^{\delta}{}_\delta{}^\epsilon + 3\eta^{\beta\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta\mathcal{J}^{\delta\alpha\epsilon} + 3\eta^{\alpha\chi}\partial_\phi\partial^\phi\partial_\epsilon\partial^\delta\mathcal{J}^{\delta\beta}{}_\delta$	
Total # constraint(s):	12		
Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{18\kappa_1^2 + 36\kappa_1^{(4)}\kappa_2^{(4)} + 18\kappa_2^2 + \kappa_7^2}{(\kappa_1 + \kappa_2)(6\kappa_1\kappa_2 + 6\kappa_2^2 + \kappa_7^2)}$
	1	0	$-\frac{1}{(\kappa_1 + \kappa_2)(6\kappa_1\kappa_2 + 6\kappa_2^2 + \kappa_7^2)}(30\kappa_2^2 + 6\kappa_2^{(4)}\kappa_7^{(4)} - \kappa_7^2 + 6\kappa_1(5\kappa_2 + \kappa_7) +$ $\sqrt{(9\kappa_1^4 + 1269\kappa_2^4 - 24\kappa_1^3\kappa_7^{(4)} + 528\kappa_2^3\kappa_7^{(4)} - 16\kappa_2^2\kappa_7^2 - 20\kappa_2^{(4)}\kappa_7^3 + 3\kappa_7^4 +$ $2\kappa_1^2(621\kappa_2^2 + 240\kappa_2^{(4)}\kappa_7 + 34\kappa_7^2) + 4\kappa_1(630\kappa_2^2 + 258\kappa_2^2\kappa_7 + 13\kappa_2^{(4)}\kappa_7^2 - 5\kappa_7^3))$
	1	0	$\frac{1}{(\kappa_1 + \kappa_2)(6\kappa_1\kappa_2 + 6\kappa_2^2 + \kappa_7^2)}(-30\kappa_2^2 - 6\kappa_2^{(4)}\kappa_7 + \kappa_7^2 - 6\kappa_1(5\kappa_2 + \kappa_7) +$ $\sqrt{(9\kappa_1^4 + 1269\kappa_2^4 - 24\kappa_1^3\kappa_7^{(4)} + 528\kappa_2^3\kappa_7^{(4)} - 16\kappa_2^2\kappa_7^2 - 20\kappa_2^{(4)}\kappa_7^3 + 3\kappa_7^4 +$ $2\kappa_1^2(621\kappa_2^2 + 240\kappa_2^{(4)}\kappa_7 + 34\kappa_7^2) + 4\kappa_1(630\kappa_2^2 + 258\kappa_2^2\kappa_7 + 13\kappa_2^{(4)}\kappa_7^2 - 5\kappa_7^3))$
Resolved unitarity condition(s):	(Demonstrably impossible)		